

ALEXANDER DORNBACK

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MS, Robotics @ Columbia | Controls, Hardware, Sim2Real | Available August 2026

EDUCATION

Columbia University MS, Mechanical Engineering , 3.95/4 Robotics, Automation, and Controls Concentration <i>Research Affiliation: Creative Machines Laboratory, Prof. Hod Lipson (January 2026 – Present)</i>	New York, NY Expected Dec 2026
Case Western Reserve University BS, Biomedical Engineering Biomechanics Concentration Minor, Music	Cleveland, OH May 2022

ENGINEERING PROJECTS

FIDO: an open-source, trainable, 3D-printed quadruped robot

- Implemented random search/hill-climbing for gait parameter optimization and PPO for sensorless joint neutral estimation
- Achieved stance in 10 minutes of parallel-environment PPO training on 4 custom linkage-drive legs in MuJoCo
- Reached stable walking in 30-minute training and completed sim-to-real pipeline for gait of 25 cm/s on real-world deployment

CATCH: Hybrid MPC-CBF impact shaping for bipedal locomotion

- Selected CBF after PID and vanilla MPC exceeded 3.34N actuator limits on impact; enforced safety across drops up to 40cm
- Built kinematic TTC vision estimator as feedforward input to dynamically adjust prediction horizon pre-impact
- At 40cm, MPC-CBF achieved 6.71N GRF vs. 8.00N for PID – a 16% reduction while ensuring hardware limit adherence

TYPHOON: A novel polar IDEX system for simultaneous multi-material 3D printing

- Designed and validated a polar dual-hotend 3D printer with simultaneous multi-material deposition and zero-purge waste
- Simulink dynamic analysis for polar axis stepper selection revealed 2-2.5x resonance amplification missed in static motor sizing

WORK EXPERIENCE

Serpent Robotics Mechatronics Intern	Philadelphia, PA June 2026 – Present
• Design a rope-suspended vertical cable-driven parallel robot using multi-winch control for autonomous in-tree positioning • Develop active tensioner architecture with PID control in embedded C++ (ESP32) on rope tension via Hall-effect dancer • Perform FMEA and Python parametric modeling on worm-gear automatic saw arm to inform motor selections • Contribute to ROS 2 migration and MuJoCo CDPR digital twin to simulate and validate advanced kinematics	
Weill Cornell Medicine Dario Institute Robotics Engineer	New York, NY Jul 2022 - Sep 2025
• Led 3 engineers over 12 months to deliver a clinically validated intubation robot, managing deployment testing with ENTs • Designed a superelastic tendon-driven actuator and transmission mechanism, achieving 30°/s tip deflection speeds • Embedded C++ (Arduino) for IR closed-loop control achieved first-pass intubation in 20-second benchmark with 90% accuracy • Developed and fabricated 3 mechanical testbeds to quantify twist, stretch, and bend of a silicone capacitive force sensor • Created an image segmentation protocol for converting 65 CT angiograms into CFD-validated multi-material 3D models	
Haima Therapeutics Formulation Scientist	Cleveland, OH Oct 2020 - Jun 2022
• Characterized 200+ nanoparticle batches, delivering foundational pre-clinical data used to secure a \$46M DARPA FSHARP grant • Co-authored peer-reviewed research: Sekhon et al. Wiley Adv. Science, 2025 (https://doi.org/10.1002/advs.202600002)	

SKILLS

- Controls: Model Predictive Control, Control Barrier Functions, PID, RL (PPO), Sim2Real
- Software: Python, ROS 2, C++, MATLAB/Simulink, I2C/SPI, Linux, Git
- Mechatronics: Embedded C++ (ESP32/Arduino/Pi Zero), Sensor Fusion, Actuator/Motor Control, Transmission Design
- Modeling and Design: MuJoCo, Gazebo, Kinematics (FK/IK), State Estimation, Forward Modeling, SolidWorks (CSWA, Sim)